

(1) Project Proposal

 Come up with coding projects of computer system simulation which can be implemented in Python...

Here are the proposed projects:

Project Name: filesystem-simulator

Programming Language: Python

Content: Create a file system simulator supporting ext4-like features including inodes, directory structures, ...

(2) Task Proposal

Project: filesystem-simulator

Task 1.1.1 Task 1.1.2 Task 1.2.1 ...

TestBlockDeviceCreation

Edge Case: Attempt to create a device with zero size or block size, or exceeding maximum size (should fail with error).

Assert that writing to one block does not affect data in other blocks.

(3) Repository Setup

 **Code Structure**

 src/  tests/

 **README.md**

 **tasks.md**

 **pyproject.toml**

 **Dockerfile**

</> Function Stubs

```
class BlockDevice:
    def __init__(self, file_path: str, total_size: int, block_size: int = 4096):
        """Default Initialization of a block device."""
        raise NotImplementedError("Block Device.__init__ not implemented")
```

 Problem Solving via API Calling

 Bug Fixing and Issue Resolution

 Test Design and Issue Reproduction

 Library Generation from Scratch

(6) Capability-Specific Adaption

(5) Functionality Implementation

 Read Test Code and Scripts

↓

 Implement Functionality Code

↓

 Verify with Provided Test Suites

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 Revise Code Implementation

(4) Unit Test Generation

2 files changed, 367 insertions(+), 0 deletions(-)
create mode 100644 tests/test_1_1_1.py
create mode 100644 tests/test_1_1_1.sh

```
root@f8d0:/workspace# bash tests/test_1_1_1.sh
collected 21 items
tests/test_1_1_1.py::TestBlockDeviceCreation::test_default_parameters FAILED
E   NotImplementedError: BlockDevice.__init__ not implemented
Some tests for Task 1.1.1 FAILED
```